

Anexo oculto

The provided dataset contains HIV-related statistics for Afghanistan (South Asia) and other UNICEF regions, spanning from 2005 to 2023. Below are the key characteristics that make it suitable for data-driven decision-making:

- This dataset is well-suited for tracking HIV trends, identifying vulnerable populations, and evaluating intervention impacts over time, provided specific questions are posed to guide the analysis.

Which age group has the highest percentage increase in new HIV cases from 2005 to 2023?"

Approach:

- $$\text{Percentage Increase} = \frac{\text{Value}_{2023} - \text{Value}_{2005}}{\text{Value}_{2005}} \times 100$$

- Analysis:**

From the dataset:

- **2005:**
 - Age 0-14, Both sexes: Value = " <0.01 " (assumed 0.005)
 - Age 15-19, Both sexes: Value = " <0.01 " (assumed 0.005)
- **2023:**
 - Age 0-14, Both sexes: Value = " <0.01 " (assumed 0.005)
 - Age 15-19, Both sexes: Value = " 0.02 "

Calculations:

- **Age 0-14:**
 - Percentage Increase = $\frac{0.005-0.005}{0.005} \times 100 = 0\%$
- **Age 15-19:**
 - Percentage Increase = $\frac{0.02-0.005}{0.005} \times 100 = \frac{0.015}{0.005} \times 100 = 300\%$

Answer:

The age group **15-19** has the highest percentage increase in new HIV cases from 2005 to 2023, with a **300% increase**, compared to the **0% increase** for the 0-14 age group.

Python Code:

```
python
... Copiar

import pandas as pd

# Simulated dataset input (subset relevant to the question)
data = {
    "Country/Region": ["Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan"],
    "UNICEF_Region": ["South Asia", "South Asia", "South Asia", "South Asia"],
    "Indicator": ["Estimated incidence rate (new HIV infection per 1,000 uninfected popu",
    "Year": [2005.0, 2005.0, 2023.0, 2023.0],
    "Sex": ["Both", "Both", "Both", "Both"],
    "Age": ["0-14", "15-19", "0-14", "15-19"],
    "Value": ["<0.01", "<0.01", "<0.01", "0.02"],
    "Lower": ["<0.01", "<0.01", "<0.01", "<0.01"],
    "Upper": ["0.01", "0.03", "0.04", "0.11"]
}

# Create DataFrame
df = pd.DataFrame(data)

# Convert "<0.01" to 0.005 for numerical analysis
df["Value"] = df["Value"].replace("<0.01", "0.005").astype(float)

# Filter for years 2005 and 2023
df_2005 = df[df["Year"] == 2005.0]
df_2023 = df[df["Year"] == 2023.0]

# Merge data for comparison
df_merged = pd.merge(df_2005[["Age", "Value"]], df_2023[["Age", "Value"]],
                    on="Age", suffixes=("_2005", "_2023"))

# Calculate percentage increase
df_merged["Percentage_Increase"] = ((df_merged["Value_2023"] - df_merged["Value_2005"])
                                   / df_merged["Value_2005"]) * 100

# Find the age group with the highest percentage increase
max_increase = df_merged.loc[df_merged["Percentage_Increase"].idxmax()]

# Output result
print(f"Age group with highest percentage increase: {max_increase['Age']}")
print(f"Percentage Increase: {max_increase['Percentage_Increase']}%")
```

Output:

```
text
... Copiar

Age group with highest percentage increase: 15-19
Percentage Increase: 300.0%
```

Notes:

- The assumption of "<0.01" as 0.005 is a simplification. If exact values were available, the percentage increase could differ, but the dataset's structure suggests 15-19 still exceeds 0-14 given the 2023 value of 0.02 vs. "<0.01".
- The analysis is limited to Afghanistan data, as it dominates the provided dataset and provides consistent age group data over the period.

